

NEEDS ANALYSIS WORKING GROUP ANALYTICAL FRAMEWORK

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BACKGROUND

The objective of the Needs Analysis Working Group (NAWG) is to conduct evidence-based and contextualized analysis of critical life-saving humanitarian needs across South Sudan to inform decision-making and prioritization by the ICCG. It aims to ensure that most vulnerable people and places are prioritised for intersectoral emergency response.

The NAWG restructure in 2022 resulted in a change in the form of output for the ICCG and the wider humanitarian community: monthly monitoring of needs severity at the county level across all counties in the country. To enable this, the NAWG analytical framework (NAF) was developed. The framework was piloted in September 2022 and was reviewed for functionality and reliability of scores/classifications of counties in December 2022-January 2023. The NAF was revised in late 2025 following consultations with NAWG members and was endorsed by the ICCG for implementation in 2026.

The NAF provides a structured framework to monitor needs and facilitate needs analysis for the monthly classification of counties and to flag locations with evidence of extreme or catastrophic humanitarian needs. The final classification (ranging from extreme to minor humanitarian needs) is an intersectoral measure of the severity of humanitarian conditions in a county, allowing for comparative analysis between counties and across time.

The framework is underpinned by analytical layers and cross-cutting components of analysis and operationalises the ICCG and HCT endorsed first-level triggers for emergency response which were agreed and endorsed by the ICCG and HCT in May 2022. The NAF was then built to get as close as possible to these indicators, given the available data in South Sudan. The indicators were as follows:

- Reports of key protection concerns faced by displaced people and other affected populations.

- Heightened conflict related sexual violence, forced recruitment of children and targeted killings of civilians
 - Deliberate destruction of civilian assets by armed actors inducing IPC 4 and above levels of food insecurity
- IPC 4 (Emergency) and 5 (Catastrophic) levels of food insecurity.
- Displaced population due to conflict or floods with the more than 20% of children and women affected by the event.
- Locations with more than 10% of IDPs compared to the host population.
- Vulnerability Trigger Criteria:
 - More than 10 % of female-headed households and other vulnerable-group headed households.
 - More than 10% presence of persons with disabilities, increased number of (male and female) youth populations requiring structured engagement.
- Disease outbreaks (including cholera, measles, meningitis, Rift Valley fever and malaria upsurges among others) as per WHO standards, disaggregated by age, sex and disability including physical type, sensory and mental.)
- Increasing/high mortality as per WHO standards, disaggregated by age, sex and disability including physical type, sensory and mental.

NAWG ANALYTICAL FRAMEWORK STRUCTURE

The analytical framework consists of five distinct steps with specific outputs at every step of analysis. These are:

1. **Underlying Vulnerability Score (UVS)** – structural vulnerability layer (0-2 points).
2. **Aggregate Trigger Score (ATS)** – shock and outcome triggers layer (0-8 points), including conflict and climate pillars. This includes the qualitative overlay where pillar scores can be adjusted based on qualitative reports shared with the NAWG team.
3. **First level Needs Severity Score (NSS)** – combined score (0-10), including methodological restrictions that place any county with IPC AFI and/or AMN > phase 4 in band A and any county with IPC AFI and/or AMN Phase 4 in minimum Band C (see methodological restrictions).
4. **NAWG contextual analysis** – discussion with members to interpret results and adjust final banding based on contextual knowledge, sectoral expertise, and additional evidence.
5. **Final Needs Severity Classification** –final severity bands agreed by NAWG members and disseminated to the ICCG and partners.

These steps are described in more detail below with explanations of each component and the indicators feeding into it.

The following process flowchart outlines the NAWG analysis components and steps.

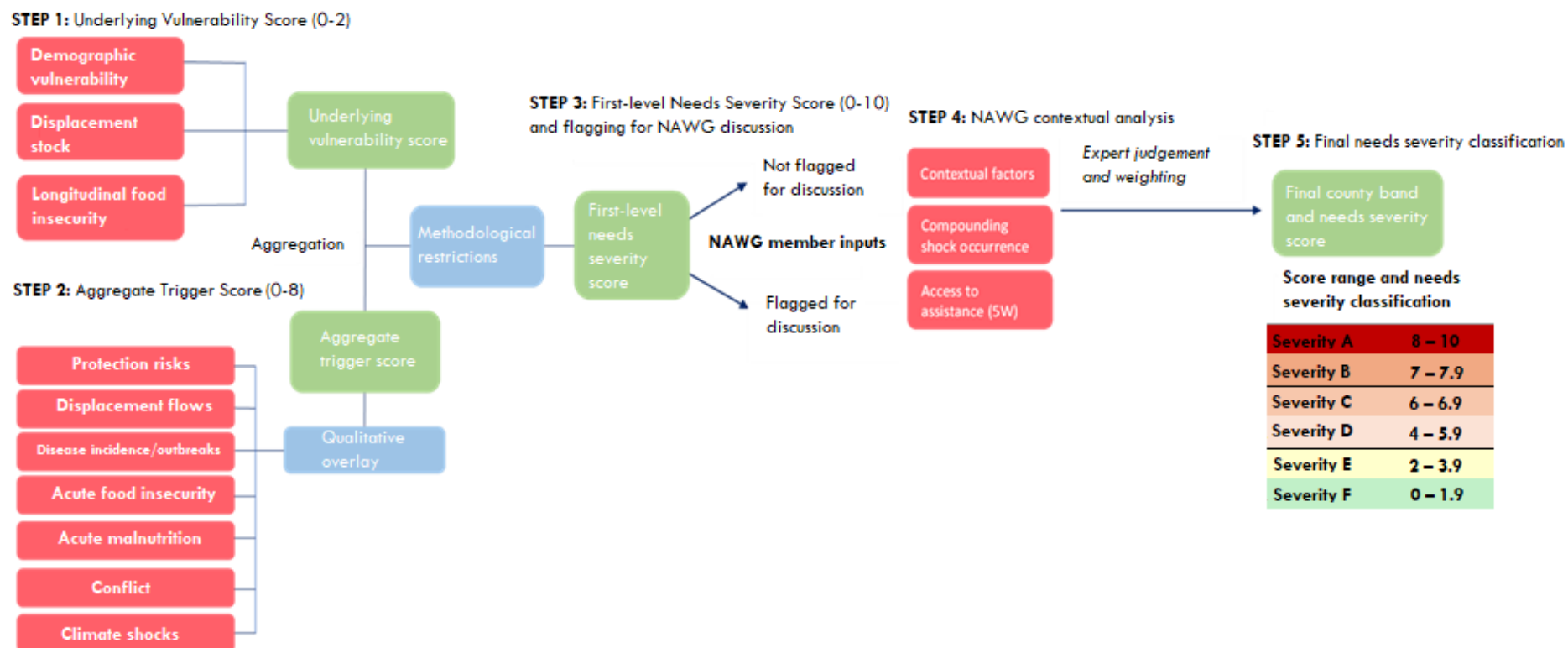


Figure 1: NAWG analysis flowchart

The indicators feeding into the UVS and the ATS are such that the **UVS accounts for 20%** of the NSS and the **ATS accounts for 80%** of the NSS:

Sub-component	Proportion of NSS
Underlying Vulnerability Score	20%
Aggregate Triggers Score	80%

The Needs Severity Scores are categorised into bands with associated definitions to describe the humanitarian situation in the county and the severity of intersectoral needs.

STEP 1: UNDERLYING VULNERABILITY SCORE (0-2)

The first step establishes the underlying (structural) vulnerability of populations in a county. The score ranges from 0 to 2, with a higher score reflecting greater structural vulnerability.

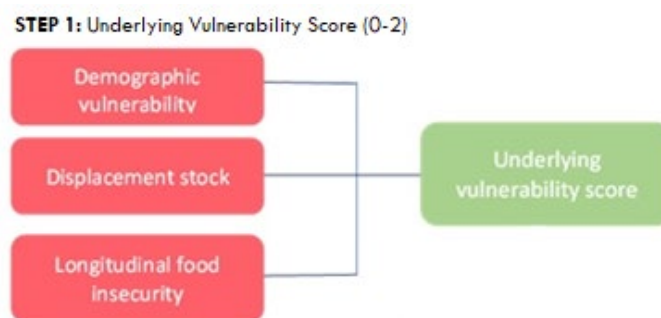


Figure 2: Underlying Vulnerability Score (UVS) components

The components and indicators in the UVS are described below, along with the data sources and the thresholds used to weight these indicators.

Table 1: Underlying Vulnerability Score (UVS) components, indicators, data sources, thresholds, weights, and rationale

Sub-component	Indicator	Data Source/temporal availability	Severity (Ranking)	Indicator Threshold	Score	Weight	Rationale/methodology
Demographic vulnerability	a. Female-headed households	FSNMS/ISNA (annual/biannual)	Very high	$x \geq p75$ (percentile based)	1 (total)	0.5	Demographic characteristics of households correlate with different aspects of needs, coping capacities and vulnerability. Two indicators based on household characteristics that impact their capacity to anticipate, cope with and recover from hazards have been selected to reflect vulnerability based on demographic profiles at the county level: (proportions of) female-headed
			High	$p50 \leq x < p75$			
			Moderate	$p25 \leq x < p50$			

			Low	$x < p25$		households and households with a person with a disability. Thresholds are revised using percentile-based cut-offs to improve granularity and reduce clustering in the highest severity category.	
	b. HHs with a person with a disability		Very high	$x \geq p75$ (percentile based)			
			High	$p50 \leq x < p75$			
			Moderate	$p25 \leq x < p50$			
			Low	$x < p25$			
	Overall demographic vulnerability (aggregate of a and b)		Very high	$x \geq p75$ (percentile based)	1		
			High	$p50 \leq x < p75$	0.75		
			Moderate	$p25 \leq x < p50$	0.5		
			Low	$x < p25$	0.25		
IDP and/or returnees (stock)	Proportion of IDPs and/or returnees in county population	IOM DTM mobility tracking (quarterly)/ISNA/FSNMS (annual/biannual)	Very high	$x \geq p75$ (percentile based)	1	0.5	Displaced and returnee populations experience heightened levels of vulnerability compared to host communities. Additionally, the presence of displaced and returnee populations potentially impacts access to food and essential services, including livelihood patterns in locations. Thresholds are revised using percentile-based cut-offs to improve granularity and reduce clustering in the highest severity category.
			High	$p50 \leq x < p75$	0.75		
			Moderate	$p25 \leq x < p50$	0.5		
			Low	$x < p25$	.25		
Longitudinal food insecurity.	IPC acute food insecurity (AFI data)	IPC (roughly (annual/biannual)	Very high	$x \geq 4$	1	1	Mean score by county between 2019-2025. The Integrated Food Security Phase Classification (IPC) conducts an annual or biannual food insecurity analysis. Scores assigned for each phase classification are as follows: Phase 1 = 1; Phase 2 = 2; Phase 3 = 3; Phase 4 = 4; and Phase 4 with pockets of 5 = 4.5
			High	$3 \leq x < 4$	0.75		
			Moderate	$2 \leq x < 3$	0.5		
			Low	$1 \leq x < 2$	0.25		

							Scores assigned to the range of phase classifications reflect the level of humanitarian priority by classification. Longitudinal analysis of these classifications provides an indication of the level of chronic vulnerability to food insecurity of populations in counties.
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STEP 2: AGGREGATE TRIGGER SCORE (0-8)

The second step quantifies shock and outcome triggers met to produce an aggregate trigger score (ATS) for each county. First-level triggers include a combination of shocks (protection risks, conflict events, climate shocks, displacement flows, and disease incidence/outbreaks) and outcomes (acute food insecurity and acute malnutrition). The more triggers and severe outcomes met in a location, the higher the ATS, ranging from 0 to 8.

The components and indicators in the ATS are described below, along with the data sources and the thresholds used to weight these indicators.

Table 2: Aggregate Trigger Score (ATS) indicators, data sources, thresholds, weights, and rationale

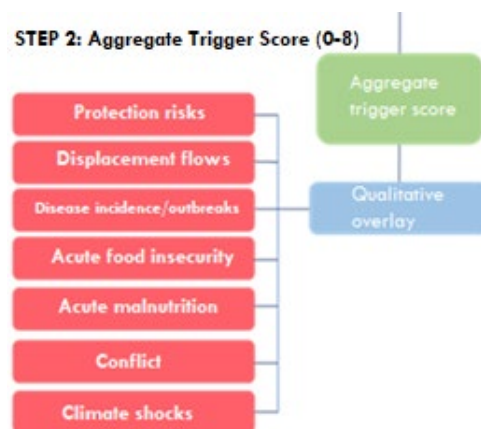


Figure 3: Aggregate Trigger Score (ATS) components

STEP 2: AGGREGATE TRIGGER SCORE (ATS) – sub-components and scoring

Sub-component	Indicator	Data Source/ temporal availability	Severity (Ranking)	Indicator Threshold	Score	Weight	Rationale/methodology
Protection severity	Protection Cluster Protection Risk Monitoring System (PRMS) combined, weighted and normalised score	Protection Cluster Protection Risk Monitoring System (PRMS)	Very high	$x > 3.36$	1	1	Indicators (and weightings) in the PRMS include abduction & kidnapping (2); attacks on civilians (3); forced family separation (2); child & forced marriage (3); discrimination & stigmatization (3); disinformation and denial on information (1); forced recruitment of child soldiers (2); gender based violence (3); restriction of access to legal process and justice (1); presence of mines and explosive ordinance (2); psychological and emotional abuse (1); theft, extortion, eviction of destruction of personal property (3); torture or cruel treatment (2); trafficking in persons, forced labour or slavery (1); restriction of movement or forced displacement (3). These indicators each range from 0 to 5 and the indicators are then weighted according to the weights in brackets and summed to give a combined score which is normalized and classified based on the thresholds given.
			High	$x > 2.36$	0.75		
			Moderate	$x > 1.36$	0.5		
			Low	$x < 1.36$	0.25		
IDP and/returnee flow	Number of displaced persons arriving and/or number of returnees returning	IOM DTM event tracking / other verifiable assessment reports (monthly)	Very high	$x \geq 10,000$	1	1	Significant numbers of arrivals of displaced persons to a community/area has the potential to exacerbate food and resource access, and to disrupt livelihood patterns.
			High	$5,000 < x < 10,000$	0.75		
			Moderate	$2500 < x < 5000$	0.5		
			Low	$0 < x < 2500$	0.25		
			None	The above conditions are not met	0		
Disease incidence/ outbreak	Disease incidence (acute watery diarrhoea (AWD), measles, acute respiratory infection (ARI), malaria) as a proportion of the total number of consultations	Health Cluster/WHO (IDSR data) (monthly)	Very high	Morbidity $x \geq 77\%$ of historic national rates	1	1	Disease outbreaks are a key driver of humanitarian needs and an outcome of other drivers of needs. They can lead to high mortality rates and be a proxy for deteriorating food security. The WHO IDSR data provides the number of reported cases of acute watery diarrhoea (AWD), measles, acute

			High	Morbidity 66% ≤ x < 77% of historic national rates	0.75		respiratory infection (ARI), and malaria from health facilities across the country. These are summed to county level and divided by the total number of consultations reported to give a proportion. These are then scored based on thresholds for each disease. This methodology is taken from the Shocks Monitoring Index that REACH used to create.
			Moderate	Morbidity 55% ≤ x < 66% of historic national rates	0.5		
			Low	Above conditions are not met	0		
	Confirmed disease outbreak	Health Cluster (monthly)	N/A	Confirmed disease outbreak = 1	1		Where reliable data on confirmed disease outbreaks is available, it is used instead of the disease incidence aggregate indicator.
Food insecurity	IPC Acute Food Insecurity (AFI) classification (current/projection) and FEWS NET classification (where available)	IPC (bi-annual) and FEWS NET (where available)	Catastrophic/ Emergency with Catastrophic pockets	IPC Food Insecurity Phase 5 or Phase 4 with pockets of Phase 5	1	1.5	The Integrated Food Security Phase Classification (IPC) conducts a biannual food insecurity analysis. After the results are published, the values for the relevant time period are used (current or projection); each is valid for 4 months. Phase 1 and 2 classifications have been aggregated here with a score of 0 assigned to reflect the relatively low humanitarian priority given to locations with these classifications. Counties with classifications of Phase 3 and above are prioritized for assistance, which is reflected in the scoring. Higher weight is assigned to Phase 5 and Phase 4 with pockets of Phase 5 locations to reflect the severity of this humanitarian needs' outcome.
			Emergency	IPC Food Insecurity Phase = 4	0.75		
			Crisis	IPC Food Insecurity Phase 3	0.5		

			Minimal/None or Stressed	IPC Food Insecurity Phase 1 and 2	0		
Malnutrition	IPC Acute Malnutrition (AMN) classifications / SMART survey data	IPC (bi-annual)/NIWG SMART surveys	Extremely Critical or Critical	IPC AMN Phase 5 Or IPC AMN Phase 4	1	1.5	The Integrated Food Security Phase Classification–Acute Malnutrition (IPC AMN) is a biannual analysis that rates the gravity of acute malnutrition in a population using SMART survey data (weight-for-height, MUAC, and the presence of bilateral oedema) and WHO emergency thresholds for Global Acute Malnutrition (GAM). Results are issued for current and projected periods, valid for 4 months. Phases 4–5 (Critical/Extremely Critical) signal life-threatening prevalence levels and receive the highest weight (score 2) in this framework. Phase 3 (Serious) indicates severe but less extreme conditions and is weighted at 0.75. Phase 2 (Alert) reflects elevated risk (score 0.5). Phase 1 (Acceptable) denotes baseline or normal conditions and receives a weight of 0. Higher weight (score 2) is assigned to critical levels to reflect the severity of this humanitarian needs' outcome.
			Serious	IPC AMN Phase 3	0.75		
			Alert	IPC AMN Phase 2	0.5		
			Acceptable	IPC AMN Phase 1	0		
Conflict	Monthly county-level conflict incident index, filtered to incident types most relevant to humanitarian impact	ACLED (monthly) and PLSO (monthly)	Very high	$x \geq p75$ (percentile based)	1	1	Conflict is a core driver of displacement, asset loss, service disruption, and access constraints. This pillar considers reliable data on conflict-related sexual violence, forced recruitment of children, targeted killings of civilians, targeted property destruction. Incidents are aggregated to monthly county level per source, standardised, and combined into a single index (using the max of standardised scores). In months where only one source is available, the pillar relies on that source. In rare months with no conflict data, the pillar is flagged as “data-missing” and handled through contextual analysis.
			High	$p50 \leq x < p75$	0.75		
			Moderate	$p25 \leq x < p50$	0.5		
			Low	$x < p25$	0.25		

Climate shocks	Monthly county-level climate shock index combining (data permitting): - Flood-related displacement - Flood extent (satellite-derived flooded area and/or population exposure estimates) - Drought stress (FAO ASI + NDVI anomaly) - Rainfall anomalies (CHIRPS 1-month anomaly)	OCHA flood displacement; satellite flood mapping (USGS/FEWS NET/Sentinel-1 or equivalent); FAO ASI; NDVI anomaly; CHIRPS rainfall anomaly	Very high	$x \geq p75$ (percentile based)	1	1	Climate shocks (floods and drought) are major drivers of displacement, livelihood loss, and acute needs in South Sudan. Sub-components are aggregated monthly at county level, converted into standardised scores using percentile-based thresholds, and combined using equal sub-weight. Counties with insufficient satellite coverage are flagged as "data-insufficient" for this pillar; contextual analysis can incorporate reliable partner information to interpret likely impact.
			High	$p50 \leq x < p75$	0.75		
			Moderate	$p25 \leq x < p50$	0.5		
			Low	$x < p25$	0.25		

STEP 3: FIRST-LEVEL NEEDS SEVERITY SCORE (0-10)

The underlying vulnerability score (0-2) and aggregate trigger score (0-8) are summed to calculate the first-level Needs Severity Score (NSS) with a possible range of 0-10. The scores are then classified into bands as per the table below. This analysis is conducted by **REACH** on a monthly basis prior to NAWG meetings.

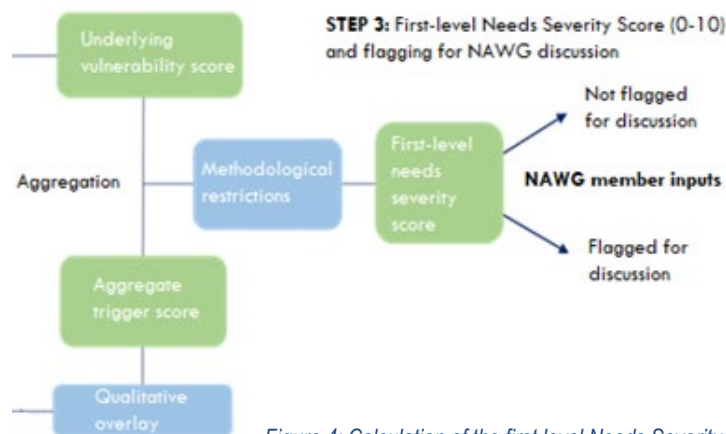


Figure 4: Calculation of the first level Needs Severity Score (NSS)

Table 3: The Needs Severity Scores are categorised into bands with associated definitions.

Needs severity classifications	Score range	Description of humanitarian conditions
Severity A	8 - 10	Extreme humanitarian conditions. Counties facing catastrophic levels of acute need across multiple sectors. Immediate, large-scale humanitarian response is critical. Populations face severe food insecurity (IPC Phase 4/5), critical malnutrition, active conflict or extreme protection threats, major disease outbreaks, and/or mass displacement. Counties in IPC AFI Phase 4 with pockets of Phase 5 are automatically placed in Band A per methodological restrictions.
Severity B	7 - 7.9	Very severe humanitarian conditions. Counties experiencing acute, multi-sectoral crises requiring urgent prioritised response. High levels of food insecurity (IPC Phase 3-4), elevated malnutrition, significant protection concerns, disease burden, and/or substantial displacement. Conditions may deteriorate rapidly without sustained intervention.
Severity C	6 - 6.9	Severe humanitarian conditions. Counties with significant humanitarian needs across multiple sectors requiring sustained response and close monitoring. Moderate to high food insecurity, concerning malnutrition trends, notable protection risks, recurrent disease outbreaks, and/or ongoing displacement pressures. Risk of escalation if conditions worsen or new shocks occur. Counties in IPC Phase 4 (food insecurity or malnutrition) are kept at minimum Band C per methodological restrictions.
Severity D	4 - 5.9	Moderate humanitarian conditions. Counties with notable but less acute humanitarian needs requiring continued monitoring and targeted response. Some food insecurity and malnutrition present but below emergency thresholds. Protection concerns and displacement exist but are more localised. Humanitarian situation is concerning but relatively more stable compared to higher bands.
Severity E	2 - 3.9	Stressed humanitarian conditions. Counties with some humanitarian concerns but limited acute needs. Food security and nutrition indicators are below crisis thresholds. Protection and displacement pressures are relatively low. Basic services are functioning, though vulnerabilities remain that could worsen under new shocks.
Severity F	0 - 1.9	None/minor humanitarian conditions. Counties with no significant acute humanitarian needs identified. Food security, nutrition, health, protection, and displacement indicators are within acceptable ranges. Basic services are largely accessible and functioning.

Note the following methodological restrictions on band classifications to reflect the severity of these outcomes and corresponding humanitarian needs:

- If a county is classified as IPC AFI Phase 4 with pockets of Phase 5 or IPC AFI Phase 5, it is automatically assigned Severity A.
- If a county is classified as IPC AMN Phase 5 (Extremely Critical), it is automatically assigned Severity A.
- If a county is classified as IPC AFI Phase 4 and/or IPC AMN Phase 4, the minimum band assigned is Severity C (6.0-6.9). Where the computed score would otherwise fall below 6.0, the score is set to 6.0 to match the minimum band.

County-level scores are shared with NAWG members¹ ahead of NAWG meeting for the following:

- a) **Inputs on the scores assigned to sub-component indicators** (protection, disease incidence/outbreak, displacement, food insecurity, malnutrition) should they assess the score/corresponding severity level does not align with their understanding/assessment of conditions and needs in the county.
- b) **Suggestions for locations to be discussed at the NAWG meeting.** Locations flagged for discussion must be based on the following criteria:
 - The first-level needs severity classification does not align with members' own assessment of the level of humanitarian needs in the county, with particular focus on:
 - a) Locations at risk of extreme/severe needs which appear to be under classified as per the first-level classifications.
 - b) Locations where the needs severity classification has changed from the previous month
 - Locations classified as having **information gaps** and/or categorized under '**close monitoring**' in a previous NAWG meeting or based on NAWG members' inputs.

A clear rationale must be provided by NAWG members when inputs are given on counties for contextual analysis in meetings.

STEP 4: NAWG CONTEXTUAL ANALYSIS

Each county flagged for the NAWG monthly meetings will undergo contextual analysis through discussion between members before the assignment of a final band based on needs severity. Additional contextual layers that may impact the level of needs severity in the county are also considered during the discussion, including:

- Contextual factors, including socio-political environment and conflict dynamics, in terms of impact on needs.
- Climate shocks, where data is available, such as UNOSAT data on populations exposed to flooding or numbers of flood-affected people.
- Cumulative shocks occurrence to examine the level and number of shocks faced by a county over a period of time and consider their impact on coping capacities.
- Access to assistance of affected populations
- Access and mobility barriers
- Any other information affecting needs severity.

NAWG members are expected to contribute to this analysis through their expertise and resources available to them. The first-level severity score and component layers of the county are evaluated against the level of needs based on context analysis and sectoral or geographic expertise. The band can then be adjusted (increased or decreased) based on the provided contextual information to reflect the assessed level of humanitarian needs in the county. Provision of evidence should be required before a band is adjusted.

¹ Members include focal points from clusters, Rapid Response Mechanism partners, UN agencies, INGOs and NNGOs.

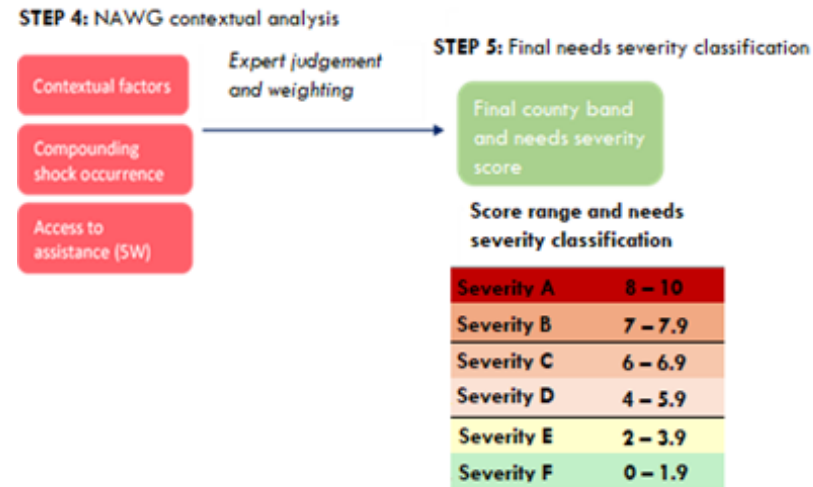


Figure 5: NAWG contextual analysis and final classification

Additional classifications:

- Close monitoring: where analysis on anticipated shocks (conflict, climate etc.) that may lead to deterioration of humanitarian needs (to Severity A or B) or trends of deteriorating humanitarian needs (to reach Severity A or B levels) merit flagging the location to the ICCG and other stakeholders to monitor closely, and for follow up in the next NAWG meeting.
- Low information county: where limited information is available to enable an understanding of humanitarian conditions and needs. To be flagged to the ICCG for follow-up/to partners for consideration for assessments. These locations will be followed up on in the next NAWG meeting.

Analysis for each county will also include (where possible) cross-cutting components of the analysis including identification of hotspots, identification of vulnerable groups and specific needs and self-prioritization of needs of affected communities, based on NAWG member inputs and where this data is available.

STEP 5: FINAL NEEDS SEVERITY CLASSIFICATION

Based on needs, expert judgment and consensus, the final band classifications are agreed upon in the NAWG meeting. These bands are detailed in Table 3.

The final Needs Severity Score (NSS) is then calculated based on the final agreed bands. When the band for a county is increased during the NAWG meeting, the score is increased to the minimum score in the new band e.g. if a county is increased to band A, the score is increased to 8.00 to match the band's minimum score. Similarly, if the band is decreased during the NAWG meeting, the score is decreased to the maximum score for the new band. e.g. if a county is decreased to band C, the score is decreased to 6.9.